Nicholas Kelly

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Objective

To advance my education and experience in Computer Engineering so that I can contribute to the field of Computer Architecture.

Experience

Jun 2023 - Present

CPU Core Architect — Ampere Computing | Portland, OR

- Micro-architecture development for core L1D, L2, and various other areas within core (e.g. MMU)
- Core-wide performance modeling (C++23) and data analysis
- Significant contributor for several infrastructure/analysis tools and flows (e.g. CI)

Jun 2016 - Jun 2023

CPU Core Architect — Intel | Hillsboro, OR

- Micro-architecture development for (inner-core) memory-system and various other areas within core (e.g. front-end)
- Core-wide performance modeling/infrastructure (C++17) and data analysis
- Primary contributor for several infrastructure/analysis tools used frequently, across teams
- Guiding adoption and education of industry-standard SW/CI practices

Jan 2015 - May 2016

Graduate Research Assistant — Prof. Mattan Erez | UT Austin | Austin, TX

- Resiliency characterization through error injection and simulation (C++, Python, Verilog)

May 2015 - Aug 2015

Validation Intern — ARM | Austin, TX

- Interconnect power and clocking validation/coverage

Qualifications

Computer Arch.

CPU simulation and various forms of data collection using C++17-23

Visualization and anaylsis in Python/Pandas/Jupyter

Scripting with Shell, Python, Ruby, Perl, and Tcl

Embedded assembly and/or C development (PIC, AVR, MSP430, ARM)

VLSI design with Verilog/SystemVerilog/UVM and various EDA tools

Software

Continuous integration with TeamCity/GitHub/GitLab, for JS/TS, Python, Ruby, and C++

Unit-test frameworks, linting, coverage, and static-analysis within JS/TS, Python, Ruby, and C++

Runtime and memory profiling of C++ programs (VTune, valgrind, jeprof, gprof)

Software-engineering practices (e.g. testing, OO, design patterns, etc.) teaching in industry

AI/ML

Usage of common LLMs in performance modeling and tool development

Development of MCP servers and skills for usage of specialized tools

Additional | Communication and support skills, across teams

Able to learn new material quickly

Education

Jan 2014 - May 2016 | **University of Texas at Austin** — Austin, TX
M.S. in Computer Architecture and Embedded Systems (3.92 GPA) - Spring 2016

Sept 2009 - Jun 2013 | **Oregon State University** — Corvallis, OR
B.S. in Electrical/Computer Engineering (3.91 GPA) - Spring 2013

Selected Projects

Jan 2014 - June 2016 | **Computer Architecture and Embedded (UT)**

- x86 (subset of ISA) processor in structural-verilog (SystemVerilog, Python, x86)
- Realtime GPU Raytracing
- Lightcuts and Illumination
- An analysis of 3DIC Kogge-stone Adders
- Auto-Multithreading extension for Node.js and V8
- GPU Power virus (genetic algorithm, code generator)
- SDF scheduling genetic algorithm to optimize towards energy usage
- Development of custom RTOS for TI Launchpad (ARM)

Sept 2012 - Jun 2013 | **VLSI/Analog Design and Simulation Projects (OSU)**

- Simulation of power-gating and near-threshold effects on power and delay for XOR gate
- Designed bike POV circuit using SystemVerilog, ModelSim, and Cadence Encounter (Place-and-route)
- Design, simulation (HSPICE/Spectre), and layout (Cadence) of OTAs for different specifications

Publications

Conferences

- Al-Otoom, D.;**Kelly, N.**; Lindsay, A.; Ahmadi, A.; Kumar, A.; Faubert, A.; Chaffin, B.; Toll, B.; Wallace, E.; Crowthers, L.; Charney, M.; Chin, M.; Spradling, M.; Witscher, S.; Caculo, S.; Fakhri, S.; Kothapalli, V.; Freelove, W.; Madhav, M., ”Performance Verification of the AmpereOne® CPU Core” In Proceedings of the 59th IEEE/ACM International Symposium on Microarchitecture® (Micro). Athens Greece. November, 2026.
- Chang, C.; Lym, S.;**Kelly, N.**; Sullivan, M. B.; Erez, M., ”Evaluating and Accelerating High-Fidelity Error Injection for HPC,” In Proceedings of The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC). Dallas, TX. November, 2018.
- Meier, R.;**Kelly, N.**; Almog, O.; Chiang, P., ”A Piezoelectric Energy-Harvesting Shoe System for Podiatric Sensing” Engineering in Medicine and Biology Society (EMBC), 2014 36th Annual International Conference of the IEEE , pp.622,625,26-30 August 2014.

Workshops

- Chang, C.; Lym, S.;**Kelly, N.**; Sullivan, M. B.; Erez, M., ”Hamartia: A Fast and Accurate Error Injection Framework,” Workshop on Silicon Errors in Logic–System Effects (SELSE). Boston, MA. April, 2018.

References available on request. Additional experience and qualifications available in alternate (e.g. web) resumes.